Lensing magnification and the thin-bin cross spectra

What magnification does to galaxy counts, what the forecast neglects by setting its coefficient to zero, and the test that measures the cost

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1 Summary

The setting. The project forecasts DESI galaxy clustering in thin redshift bins – angular power spectra C_ℓ^{gg}, computed without the Limber approximation, joint with LSST Y1 cosmic shear – with Fisher matrices: the forecast's error bars come from the derivatives of the predicted data at one fiducial point. That machinery is built step by step in `notes/fisher_machinery`; this note needs only its vocabulary. The data vector includes cross spectra between different lens bins, and the companion note `notes/cross_correlations` shows that extending the measured pair list raises FoM(w_0, w_a) by up to 55–59% on the three thinner layouts.

The question. Light from a galaxy is deflected by the matter in front of it, and the deflection changes how many galaxies a survey counts in each direction – an effect whose standard name, used once here and then replaced for the reasons below, is *magnification bias*. In this project "bias" otherwise always means galaxy bias (the ratio of galaxy to matter clustering), and a displacement of a recovered parameter is always a "shift", never a bias. So this note names the effect once, and from here on speaks of the *magnification coefficient* C_i of lens bin i (the number that multiplies the lensing convergence in the observed counts, Section 3) and of the *count slope* s (the property of the sample that sets C_i). The forecast currently sets $C_i = 0$ in every lens bin. The question, raised by the cross-correlation study: once more cross spectra are measured, does the model need the magnification terms – and does the answer change between the paper's pair lists and the extended ones?

What is known without new runs. Three ingredients are already in hand. (i) *The literature.* DES Y3 modelled exactly the term this project switches off, measured its coefficients, and found that neglecting it shifts S_8 by up to 0.85σ for its wider, deeper lens sample [5, 4]; Euclid's spectroscopic forecast finds 0.2–0.7σ shifts from neglecting it even in a thin-bin analysis without cross spectra, at higher redshift and larger coefficients than DESI's [8], and the neglected term can also shift n_s and the neutrino-mass sum [11]; a dedicated spectroscopic study finds the effect almost irrelevant for DESI's samples because their coefficients are small, while flagging that the coefficients need better modelling [7]. (ii) *The code audit.* Cosmolike implements the DES Y3 model – one constant coefficient per lens bin times the bin's convergence – correctly in its auto-spectrum paths, Limber and non-Limber; the one hard restriction was that the gg cross-pair engine refused a nonzero coefficient (`cosmo2D.c:13115`), keeping magnification off exactly the spectra where it matters most (Section 5). (iii) *The pipeline numbers.* The restriction of (ii) has since been lifted: the cross-pair engine now carries the magnification rows, validated against the autos and the existing vectors (Section 5.6), and the shift test ran at the final derivative tier on every layout and both pair lists (Section 6). An earlier external Limber estimate, recorded in Section 4.5, came out about ten times low on the

thin layouts and did not cover $\Delta z = 0.05$ and $\Delta z = 0.2$; every conclusion below rests on the pipeline numbers.

The answer in numbers. Every shift below is a what-if number – if nature carries magnification at the stated coefficients and the model leaves it out, where does the fit land, in units of the forecast's own errors – with no threshold line (Section 6, Figure 3). At the measured coefficients (BGS $C = 2$, LRG $C = 3$, ELG 0), the thin-layout fits move by at most 0.07 of the error, on both pair lists: at those coefficients the thin-bin forecasts, paper pairs or the extended "pairmax" pairs (Section 2), lose essentially nothing by leaving magnification out. The unmeasured ELG count slope dominates the uncertainty: at the bracket corner $C_{\rm ELG} = +2$ the pairmax lists reach 0.30–0.46, and there the free nuisances *amplify* the shift rather than absorb it – the magnification excess on the added pairs reads as a change of the per-group linear-bias level and slope, the same freedom that produces the pairmax FoM gain, so the gain and the magnification sensitivity travel together. At $C_{\rm ELG} = -2$ the paper lists themselves reach 0.2–0.3. The wide $\Delta z = 0.2$ layout is magnification dominated – its measured adjacent cross spectra change by up to 25 times their magnification-free size – and moves 0.19 even at the measured coefficients, with A_s shifting by about 1σ at $C_{\rm ELG} = +2$. Measuring the ELG count slope (Section 3.7) is therefore the study's single most valuable next step.

2 Fiducial model and parameter values

Every number of this note is computed at the fiducial below, the evaluate point – the parameter values at which the data vectors and derivatives are evaluated – of the paper configurations `EXAMPLE_EVALUATE1`–`4.yaml`, frozen here so the note does not depend on files that may change.

Cosmology. Flat $w_0 w_a$CDM: $10^9 A_s = 2.1$ (A_s the amplitude of the primordial curvature power spectrum), $n_s = 0.96605$, $H_0 = 67.32$ km/s/Mpc, $\Omega_b = 0.04$, $\Omega_m = 0.3$ (neutrinos included), $w_0 = -1$ and $w_a = 0$; the neutrino mass sum is 0.06 eV and the optical depth $\tau = 0.0697$; derived $\sigma_8 = 0.8278$. The Fisher matrices vary $10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0 and w_a with flat priors.

Layouts, bands and pair lists. Four lens-bin layouts: $\Delta d_{\rm com} = 62$ Mpc (76 bins), $\Delta z = 0.02$ (90), $\Delta z = 0.05$ (36) and $\Delta z = 0.2$ (10); every spectrum is measured in 53 band powers of width $\Delta\ell = 10$ over $15 \le \ell \le 544$. The *paper pair lists* keep the cross pairs whose blurred redshift distributions overlap by at least 10^{-6} at the fiducial velocity dispersions (`data/<stem>_pairs.txt`): 107, 133, 52 and 13 pairs on the four layouts. The *pairmax lists* of the pair-axes study (378, 370, 178 and 32 pairs) add deeper pairs, with a reach that differs by layout. Writing g for the distance between the two bins' raw redshift ranges in bin widths – the convention of `notes/cross_correlations`, in which an adjacent pair has $g = 0$ and $C_\ell^{i,i+2}$ has $g = 1$ – the pairmax lists keep $g \le 3$ on $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.05$, $g \le 2$ on $\Delta z = 0.2$, and on $\Delta z = 0.02$ the pairs with $g \le 2$ plus those of the $\sigma_v = 450$, 600, 750, 900 and 1200 km/s overlap lists (the overlap criterion rerun with every group's dispersion set to that one value). How far apart the listed pairs are *in distance* varies strongly by layout (Section 4.2, Table 2): the thin layouts' lists are radially local, the $\Delta z = 0.2$ list is not. The forecast sky is 12,300 deg^2; the covariance is Gaussian, built at 14,000 deg^2 with its inverse rescaled by 12,300/14,000.

Galaxy bias. The group amplitudes are $B_g = 1.34$ (BGS), 1.7 (LRG1–LRG3) and 0.84 (ELG1, ELG2); at the fiducial every tilt and residual is zero, so bin i has linear bias $b_{1,i} = B_g/D(\bar{z}_i)$, with the growth factor D ($D = 1$ today) read off the linear power spectrum at $k = 0.005$ Mpc^{-1} and $\bar{z}_i$ the bin's mean redshift. The one-loop terms are those of the fiducial model: quadratic and tidal biases on their calibrations, $\alpha_2 = 0$ in every bin, the stochastic amplitudes zero, the pessimistic linear-bias prior, and every velocity-dispersion prior width six times its base width (`notes/bias_model_summary`).

Velocity dispersions. Seven dispersion groups with fiducials 371, 358, 357, 348, 343, 304 and 289 km/s (BGS split at $\bar{z} = 0.255$); each thin bin's redshift distribution is convolved with the group's Gaussian blur before any kernel is built.

Scale cuts. The non-Limber ultraviolet-fraction scale cut [18]: bin i is cut at $k_{\max}/D(\bar{z}_i)$, and a band is kept while less than 5% of it comes from the power spectrum above that cut, intersected with the shipped mask ($k = 0.1185$ Mpc^{-1}, about $0.176\,h$/Mpc, in every bin). As everywhere in this project, a quoted $k_{\max}$ is the value at $z = 0$.

Magnification coefficients. The fiducial coefficients adopted for the tests, with the measured count slopes behind them (defined and derived in Section 3; sources in Table 1): BGS $C = +2.0$, LRG $C = +3.0$, ELG bracketed in $C \in [-2, +2]$ with 0 as the central value. One constant per sample group, assigned to every bin of the group.

3 What lensing magnification is

This section builds the effect from the ground up. Nothing in it is specific to this project; everything downstream depends on it.

3.1 The magnifying-glass picture

Light travels to us through a lumpy universe. Where the line of sight passes more matter than average, the extra gravity bends the rays toward the mass, exactly as a glass lens bends rays toward its axis. A patch of sky behind an overdense foreground therefore behaves as if seen through a weak magnifying glass: everything in the patch looks slightly *larger* and, because the bending conserves surface brightness (photons are neither created nor destroyed, only redirected), slightly *brighter*. Behind an underdense foreground the same logic runs in reverse: a weak reducing glass. The magnification varies from direction to direction, tracing the foreground matter – and that is the whole effect. Everything below is bookkeeping of what a magnifying glass does to galaxy counts.

3.2 Convergence, shear, and $\mu \simeq 1 + 2\kappa$

Two numbers describe the distortion of a small patch of sky. The *convergence* κ is the isotropic part: positive κ enlarges images in all directions equally. The *shear* γ is the anisotropic part: it stretches images along one axis and squeezes them along the other, turning circles into

ellipses, and is the quantity cosmic-shear surveys measure. Formally, the mapping from true to observed positions has the Jacobian matrix

$$A = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}, \tag{1}$$

whose determinant is the ratio of true to observed solid angle. The *magnification* μ is the ratio of observed to true solid angle, the inverse determinant:

$$\mu \;=\; \frac{1}{\det A} \;=\; \frac{1}{(1-\kappa)^2 - |\gamma|^2} \;\simeq\; 1 + 2\kappa, \tag{2}$$

where the last step expands to first order in κ and γ, the weak-lensing limit ($\kappa, |\gamma| \ll 1$): the shear enters μ only at second order, so to first order magnification is set by the convergence alone. For scale: along a typical line of sight to $z \sim 1$, κ is of order 10^{-2} at most, so $\mu - 1 \lesssim 2\%$.

Physically, κ in a given direction is the matter overdensity along the line of sight, weighted by how efficiently each distance lenses (Section 4.1 derives the weight):

$$\kappa_j(\hat{\mathbf{n}}) = \int_0^{\chi_j} d\chi \, W_{\mathrm{mag},j}(\chi) \, \delta_m(\chi\hat{\mathbf{n}}, \chi), \tag{3}$$

with δ_m the matter density contrast, χ the comoving distance, and $W_{\mathrm{mag},j}$ the lensing-efficiency kernel of the galaxies of bin j. The subscript j matters: each lens bin has its own convergence, sourced by the matter in front of *its* galaxies.

3.3 The two competing effects on counts

A survey counts galaxies per unit solid angle above a flux limit. Magnification changes both the solid angle and the fluxes (Figure 1):

1. **Area dilution.** A patch of true solid angle $\Delta\Omega$ is observed with solid angle $\mu\,\Delta\Omega$: the same galaxies are spread over more sky, so the observed number density *drops* by the factor μ. With Eq. (2), that is a contribution -2κ to the observed density contrast.

2. **Flux brightening.** Each galaxy's image area grows by μ while its surface brightness (flux per unit image area) is conserved, so its total flux grows by μ. Galaxies that sat just below the survey's flux limit are lifted *into* the sample, so the counts *rise* – by an amount set by how many galaxies sit just below the limit, i.e. by the steepness of the counts at the limit.

Which effect wins depends on the sample, through one number, defined next.

3.4 The count slope s

Let $N(< m)$ be the cumulative number of sample galaxies per unit solid angle brighter than apparent magnitude m (recall that larger m means fainter: $m = -2.5\log_{10} F + \mathrm{const}$, with F the flux). The *count slope* at the sample's magnitude limit m_{cut} is

$$s \;\equiv\; \left.\frac{d\log_{10} N(<m)}{dm}\right|_{m_{\mathrm{cut}}}. \tag{4}$$

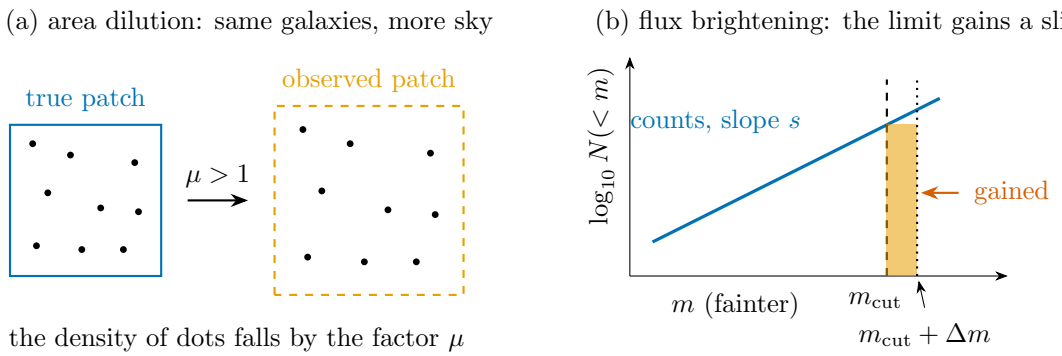


Figure 1: The two competing effects of a magnification $\mu > 1$ on galaxy counts. (a) Area dilution: the observed patch (dashed orange) covers μ times the solid angle of the true patch (solid blue) and holds the same galaxies, so the number density falls by μ. (b) Flux brightening: each galaxy brightens by μ, which shifts the effective magnitude limit faintward by $\Delta m = 2.5 \log_{10} \mu$; the sample gains the galaxies in the shaded sliver, a fraction set by the count slope s (the slope of the blue line at the cut). The net effect is the difference of the two.

It answers: if the survey could see 0.1 magnitude deeper, by what fraction would the sample grow? (By $10^{0.1s} - 1 \approx 0.23\, s$.) A steep luminosity function near the cut gives large s; a sample whose counts have flattened near the cut (a bright, nearly complete sample) gives small s. An equivalent convention uses the counts as a function of flux, $n(> F) \propto F^{-\alpha}$; since $F \propto 10^{-0.4m}$, the two slopes are related by

$$\alpha = 2.5\, s. \tag{5}$$

3.5 The derivation of $\delta_\mu = (5s - 2)\,\kappa$, step by step

Work in one direction $\hat{\mathbf{n}}$ with magnification μ, and compare the observed number density $n_{\rm obs}$ (per observed solid angle, above the observed flux limit $F_{\rm cut}$) with the density n_0 that would be observed with no lensing.

Step 1: what magnification does to the inputs. Fluxes: every galaxy's observed flux is μ times its unlensed flux, so a galaxy passes the cut if its unlensed flux exceeds $F_{\rm cut}/\mu$. Solid angle: a patch holding the galaxies has observed solid angle μ times its true solid angle.

Step 2: the observed density. Therefore

$$n_{\rm obs}(> F_{\rm cut}) = \frac{1}{\mu}\, n_0\!\left(> \frac{F_{\rm cut}}{\mu}\right) : \tag{6}$$

the $1/\mu$ in front is the area dilution; the deeper effective limit $F_{\rm cut}/\mu$ is the brightening.

Step 3: expand the counts at the limit. Near the cut the counts behave as a power law, $n_0(> F) \propto F^{-\alpha}$ with α the flux slope of Eq. (5). Then

$$n_0\!\left(> \frac{F_{\rm cut}}{\mu}\right) = \mu^{\alpha}\, n_0(> F_{\rm cut}), \qquad \text{so} \qquad n_{\rm obs} = \mu^{\alpha - 1}\, n_0. \tag{7}$$

Step 4: linearize. With $\mu \simeq 1 + 2\kappa$ from Eq. (2) and $\kappa \ll 1$,

$$\frac{n_{\rm obs} - n_0}{n_0} \;\equiv\; \delta_\mu \;=\; (\alpha - 1)\, 2\kappa \;=\; (5s - 2)\, \kappa. \tag{8}$$

The same arithmetic in magnitude language, as a check: the brightening shifts every magnitude by $\Delta m = -2.5 \log_{10} \mu \simeq -(5/\ln 10)\, \kappa$, so the sample gains a fraction $\ln 10 \cdot s \cdot |\Delta m| = 5s\,\kappa$ (the sliver of Figure 1b), while the dilution removes 2κ; the net is $(5s - 2)\, \kappa$.

The worked toy: $s = 0.4$ cancels. Take $\kappa = 0.01$, so $\mu = 1.02$: the patch grows by 2% and every galaxy brightens by 2% ($\Delta m \simeq -0.0215$ mag). For a sample with $s = 1$ (the DESI LRGs, Table 1), the brightening pulls in $5s\kappa = 5\%$ more galaxies while the dilution costs 2%: the counts rise by 3%, i.e. $C = +3$. For a sample with $s = 0.4$ the gain is $5 \times 0.4 \times 0.01 = 2\%$, exactly the dilution: the counts do not change at all, however strong the lensing. For a bright, flat-counts sample with $s = 0.2$ the gain is only 1% and the counts *drop* by 1%: behind an overdensity such a sample looks emptier, $C = -1$.

3.6 The conventions map

One effect, several notations in the literature and the code. All of the following are the same number:

$$C \;=\; 5s - 2 \;=\; 2(\alpha - 1) \;=\; \texttt{b_mag} \text{ (cosmolike)}. \tag{9}$$

DES Y3 [4, 5] writes $\delta_\mu = C\,\kappa$ and splits $C = C_{\rm sample} + C_{\rm area}$ with $C_{\rm area} = -2$ and $C_{\rm sample}$ the measured response of the selection ($C_{\rm sample} = 5s$ for a pure magnitude cut; their general definition also covers size selection, adding a size-slope term $\partial \ln n / \partial \ln r$ at the cut [4]). Cosmolike's per-bin nuisance parameter `b_mag` multiplies the convergence kernel directly, so it is the *total* coefficient C, area term included – any wiring of measured slopes must set $\texttt{b_mag} = 5s - 2$, never s. In this note the symbol is C_i, the magnification coefficient of lens bin i.

3.7 How s is measured for a real selection

For a pure magnitude-limited sample, s can be read off the luminosity function. Real selections are not pure magnitude cuts – they combine colours, total magnitudes, fibre magnitudes and sizes – and then the only reliable method is the *finite difference*: take the target catalog, shift every magnitude by a small $\delta m = \pm 0.01$ (the response of each quantity to magnification applied individually), re-run the full selection, and difference the counts [5, 15, 14]. Two subtleties move the answer by tens of percent:

- **Fibre magnitudes respond less than total magnitudes.** A fibre magnitude measures the flux inside a fixed aperture. A magnified galaxy is brighter *and larger*, so a smaller fraction of its flux falls inside the aperture: the fibre flux rises by less than μ. A selection with fibre-magnitude cuts (DESI's LRG and ELG selections have them) therefore responds less than the total-magnitude arithmetic suggests. Ignoring this overestimates the DESI LRG slope by $\sim$13% for the full sample and $\sim$30% in its faintest bin [15] – and the signal scales with $s - 0.4$, which amplifies the error further.

sample	count slope s	adopted $C = 5s - 2$	source
BGS	0.81, 0.80 (two bins, $z_{\rm eff} = 0.21,\ 0.35$)	+2.0	finite diff. [17]
LRG	0.97–1.04 per bin; 1.008 ± 0.007 full	+3.0	finite diff. [15]
	0.999 ± 0.015 (DECaLS, DESI-like)		also [16]
ELG	no measurement; models only	0, bracket $[-2, +2]$	[7]

Table 1: Measured count slopes of the DESI samples and the magnification coefficients this study adopts (one constant per sample group). The LRG and BGS slopes include the fibre-flux and photometric-redshift responses (Section 3.7). The ELG bracket spans the plausible range of a faint, colour-selected sample; the finite-difference measurement on the DESI ELG selection is an open item.

- **Size cuts respond with the opposite sign.** A cut that rejects large images (star–galaxy separation does) loses galaxies when magnification enlarges them, partially cancelling the flux gain. DES Y3 found its image-simulation estimates (which carry the size response) systematically below its flux-only estimates for this reason [5].

Within one sample the slope also drifts with redshift – by up to a factor of three across the BOSS CMASS sample [14] – which is why a per-bin coefficient (Section 5) is the right level of modelling for thin bins.

3.8 The measured DESI values

Table 1 collects the measured count slopes for the DESI samples and the coefficients adopted by this study. The LRG and BGS slopes are finite-difference measurements on the DESI target catalogs; no equivalent measurement exists for the ELGs, whose published values are luminosity-function models only [7], so the study brackets $C_{\rm ELG}$ and treats the measurement as an open item (Appendix A).

Pending – T0 (optional): finite-difference count slopes per thin bin for BGS, LRG and ELG from the DESI target selections, with the fibre-magnitude response; one figure $s(z)$ per sample. Until it runs, Table 1 holds group constants and the ELG bracket.

4 Why cross spectra see it most

4.1 The lensing efficiency, step by step

How strongly does matter at distance χ lens a source at distance χ_s? Three factors build the answer.

Step 1: the deflection. A lump of matter at χ bends a passing ray by an angle proportional to the lump's mass – more precisely, to the transverse gradient of its gravitational potential. Through Poisson's equation that gradient traces $\Omega_m\, \delta_m / a$, with a the scale factor at the lens (the $1/a$ because the potential of a fixed comoving overdensity grows as the universe expands around it).

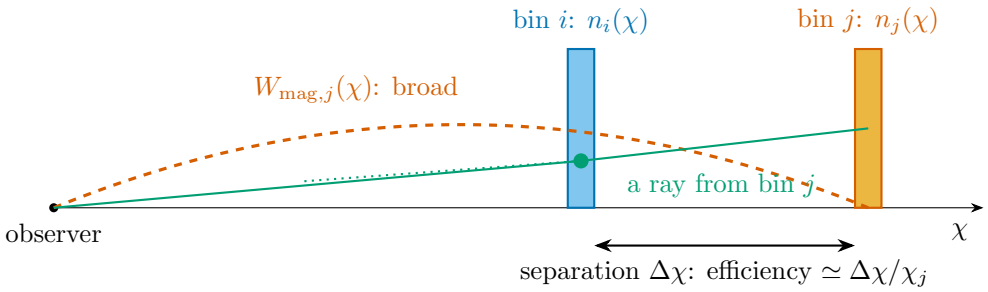


Figure 2: The geometry of lens-bin magnification. The density kernels of two thin bins (solid boxes) are narrow and disjoint, so their density cross spectrum is tiny. The magnification kernel of the farther bin (dashed curve, Eq. (10)) is broad: it covers the whole foreground, bin i included, so the matter that bin i's galaxies trace also magnifies bin j's counts (green ray). At bin i's location the kernel's height is the lensing efficiency $\simeq \Delta\chi/\chi_j$ times $(1+z_i)\,\chi_i$ (Eq. (11)): small for close pairs, large for far ones.

Step 2: the geometry. A fixed deflection at the lens displaces the *apparent* source position by an angle proportional to $(\chi_s - \chi)/\chi_s$: a deflection right at the source ($\chi = \chi_s$) moves its image not at all, a deflection halfway moves it most effectively relative to the lever arms. One more factor χ converts the angular scale of the lensing mass to a transverse physical scale. The product $\chi\,(\chi_s - \chi)/\chi_s$ is the famous lensing-efficiency triangle: zero at the observer, zero at the source, broad maximum halfway in between.

Step 3: average over the bin. The galaxies of lens bin j are distributed in distance as the bin's (blurred) $n_j(\chi)$, so the kernel of Eq. (3) averages the triangle over the bin:

$$W_{\mathrm{mag},j}(\chi) = \frac{3\Omega_m H_0^2}{2c^2}\,\frac{\chi}{a(\chi)}\int_\chi^\infty d\chi'\; n_j(\chi')\,\frac{\chi'-\chi}{\chi'}. \tag{10}$$

For a thin bin at χ_j the inner integral is simply $(\chi_j - \chi)/\chi_j$. The kernel is *broad*: it extends from the observer to the bin, whatever the bin's width. That breadth, against the thinness of the density kernels, drives everything in this note (Figure 2).

4.2 Hand-worked efficiencies and kernel heights

Two numbers describe how strongly a pair (i, j), $\chi_i < \chi_j$, couples density to magnification. The first is the *efficiency* of Eq. (10) at the nearer bin, $(\chi_j - \chi_i)/\chi_j = \Delta\chi/\chi_j$. The second – the one that actually sets the cross term – is the *kernel height*, the full kernel at bin i:

$$W_{\mathrm{mag},j}(\chi_i) = \frac{3\Omega_m H_0^2}{2c^2}\,(1+z_i)\,\chi_i\,\frac{\Delta\chi}{\chi_j} \;\equiv\; \frac{3\Omega_m H_0^2}{2c^2}\,h_{ij}, \qquad h_{ij} = (1+z_i)\,\chi_i\,\frac{\Delta\chi}{\chi_j}. \tag{11}$$

For radially close pairs $\chi_i \approx \chi_j$, so $h_{ij} \approx (1+z_i)\,\Delta\chi$: pairs with the same radial separation have nearly the same kernel height wherever they sit. A low-redshift pair can therefore carry a large *efficiency* (χ_j is small) and still a modest kernel: the small $(1+z_i)\,\chi_i$ gives back what the efficiency gained.

Worked cases, by hand, quoting h_{ij} in Mpc (the prefactor is common to all). Pair depth uses the g convention of Section 2 (adjacent = $g = 0$). The thin $\Delta z = 0.02$ bins are ≈ 55 Mpc wide; near the ELG range $\chi \approx 3200$ Mpc (the distance the cross-correlations note's toy uses at $z = 0.93$):

- **Adjacent pair, $g = 0$ (the paper lists):** centres one bin width apart, $\Delta\chi \approx 55$ Mpc: efficiency $\approx 55/3255 \approx 0.017$, $h \approx 1.92 \times 3200 \times 0.017 \approx 100$ Mpc.
- **Added pair, $g = 2$ (the pairmax lists):** centres three widths apart, $\Delta\chi \approx 165$ Mpc: efficiency ≈ 0.049, $h \approx 300$ Mpc. The deepest thin-layout additions, $g = 3$ (centres four widths apart, $\Delta\chi \approx 220$ Mpc), reach efficiency ≈ 0.064 and $h \approx 390$ Mpc.
- **A far pair (in no current list):** BGS at $z = 0.3$ ($\chi \approx 1250$ Mpc) against an ELG bin at $z = 0.92$: efficiency $\approx (3200 - 1250)/3200 \approx 0.61$, ten times the $g = 3$ value – but $h \approx 1.30 \times 1250 \times 0.61 \approx 990$ Mpc, only 2.5–3 times the added pairs' heights: the small $(1 + z_i)\,\chi_i$ at low redshift gives back most of the efficiency gain. The measured totals of Table 3 fold the coefficients, biases and band errors on top of these heights.

The same offset defuses the largest paper-list efficiencies of the thin layouts, which sit at their lowest-redshift BGS adjacent pairs: on $\Delta z = 0.02$ the pair $\bar{z} = 0.110 \times 0.130$ has $\Delta\chi/\chi_j \approx 0.15$ – the largest on that paper list – but $h \approx 1.11 \times 480 \times 0.15 \approx 80$ Mpc, below the ELG adjacent pair's ≈ 100 Mpc.

Table 2 gives the efficiencies of every listed pair, evaluated at the bin mean redshifts. The layouts differ in kind. The thin layouts' paper lists are radially local (median efficiencies 1.5–4%, maxima at the defused low-z BGS pairs), and their pairmax lists reach 0.34–0.59 only at single deep pairs. On $\Delta z = 0.2$ *every* pair is far apart in this sense: its adjacent BGS pair ($\bar{z} = 0.203 \times 0.345$) has efficiency 0.39 and $h \approx 1.20 \times 860 \times 0.39 \approx 400$ Mpc – the kernel height of a thin-layout $g = 3$ pair, on a *measured* paper pair – and even its cross-sample LRG2 × ELG1 pair ($\bar{z} = 0.706 \times 0.906$) has efficiency 0.18, where the thin layouts' cross-sample pairs sit at essentially the same redshift with $\Delta\chi \approx 0$. The $\Delta z = 0.2$ layout is the most DES-like – wide bins, large separations, and by the mode counting of Section 4.4 the largest relative weight of magnification per spectrum; the DES lens bins that made magnification a headline systematic span the survey depth with efficiencies of order one half [5]. Sections 4.5 and 6 confirm what this geometry predicts: the wide layout's cross spectra are magnification dominated, and its shifts are the largest of the four layouts.

4.3 Disjoint bins: density vanishes, magnification does not

In the Limber approximation the density cross spectrum of two bins multiplies the two density kernels at the same distance, $\int d\chi\, W_i W_j \ldots$, which is exactly zero when the (blurred) supports do not overlap; the exact non-Limber value is small but finite, carried by long radial modes (`notes/cross_correlations`, "Why bins with zero overlap still correlate"). The density×magnification term needs no overlap at all: the magnification kernel of the farther bin covers the nearer bin's support entirely (Figure 2). So as a pair's separation grows, the density term dies quickly while the magnification term *grows* (the efficiency rises with $\Delta\chi$): far enough apart, magnification is the whole signal. DES Y3 saw exactly this – its widest-separated lens-bin pair is magnification dominated [5] – and the cross-correlation of disjoint DESI photometric bins detects the magnification signal at 11σ [17]. On the thin layouts' radially local pair lists

layout	paper list: median / max	pairmax list: median / max
$\Delta d_{\rm com} = 62$ Mpc	0.020 / 0.120	0.056 / 0.351
$\Delta z = 0.02$	0.015 / 0.149	0.032 / 0.342
$\Delta z = 0.05$	0.038 / 0.271	0.094 / 0.589
$\Delta z = 0.2$	0.132 / 0.390	0.250 / 0.672

Table 2: The lensing efficiency $\Delta\chi/\chi_j$ of every pair of the paper and pairmax lists, evaluated at the bin mean redshifts for flat ΛCDM with $\Omega_m = 0.3$ and $h = 0.6732$: the median and the maximum per list. The thin layouts' paper lists are radially local; the $\Delta z = 0.2$ lists are not. The efficiency alone overstates the low-redshift pairs: the kernel height of Eq. (11) carries the extra factor $(1 + z_i)\,\chi_i$, which defuses them (Section 4.2).

the crossover is never reached; the $\Delta z = 0.2$ lists, whose pairs are genuinely far apart (Table 2), are *past* it – the pipeline finds their measured adjacent cross spectra magnification dominated, with $|\delta d/d|$ up to 25 (Section 4.5).

4.4 Mode counting: why thin bins do not amplify magnification

Thin bins are this project's defining choice, and they work for the density because of mode counting: a bin of width w averages the density over w, so halving w doubles the density power per spectrum ($C_\ell^{\rm DD} \propto \int W_i^2 P/\chi^2 \sim P/(w\chi^2)$ for a normalized top hat) and resolves more radial modes. The magnification terms do not share the gain: in $\int d\chi\, W_i\, W_{{\rm mag},j}\, P/\chi^2$ the normalized thin kernel simply samples the broad kernel at the bin, *independently of* w, and the magnification×magnification term has no thin kernel at all. Thinning the bins therefore raises the density signal and leaves the magnification terms fixed: their *relative* weight per spectrum falls. The Euclid photometric magnification study states the same conclusion from the other direction: fewer, wider bins make magnification matter more; many thin bins boost only the density and RSD modes [9]. This is one of the few places where the thin-bin design actively protects the forecast.

4.5 The size of the neglected signal

With the extended code (Section 5.6), the size of the neglected signal is measured directly: $\Delta\chi^2 = \delta d^T \mathsf{C}^{-1} \delta d$, with δd the magnification part of the data vector – the difference between the vector with and without the coefficients – computed with the blurred $n(z)$, the real bands and masks, and C the forecast covariance. Table 3 gives the totals on the shipped mask at the fiducial coefficients and at the ELG bracket corners.

The $\Delta z = 0.2$ regime: magnification-dominated cross spectra. On the wide layout, magnification changes the *measured* adjacent cross spectra by large factors – $|\delta d/d|$ reaches 25 on the pair of bins 4 and 5 – while on the thin layouts the measured pairs change at the percent level. Section 4.2 supplies the reason from both sides of the ratio. The numerator is large: a $\Delta z = 0.2$ adjacent pair spans $\Delta\chi$ of several hundred Mpc, so its magnification kernel height is $h \approx 400$ Mpc, that of a deep thin-layout pair, on every one of its measured pairs. The denominator is small: two wide adjacent bins overlap only through the thin blurred sliver at their shared edge, so their density cross spectrum is a small residual, while two thin adjacent

	paper list			pairmax list		
layout	$C_{\rm ELG} = 0$	$+2$	-2	$C_{\rm ELG} = 0$	$+2$	-2
$\Delta d_{\rm com} = 62$ Mpc	0.55	1.30	8.81	2.61	6.64	26.8
$\Delta z = 0.02$	0.48	0.88	7.68	1.44	3.02	18.0
$\Delta z = 0.05$	1.55	3.07	18.8	12.9	22.6	76.7
$\Delta z = 0.2$	15.8	39.3	82.4	63.1	188	404

Table 3: The pipeline $\Delta\chi^2$ of the neglected magnification signal on the shipped mask ($k = 0.176\,h$/Mpc in every bin), per layout and pair list, at the fiducial coefficients (BGS $C = 2$, LRG $C = 3$, $C_{\rm ELG} = 0$) and at the two ELG bracket corners. $\Delta\chi^2$ bounds a perfectly aligned shift at $\sqrt{\Delta\chi^2}$; the actual shifts, after projection onto the parameter directions, are in Section 6. On the paper scale cuts at $k_{\rm max} = 0.2\,h$/Mpc (at $z = 0$) the fiducial-coefficient totals are smaller: 0.54 / 0.47 / 1.25 / 11.6 on the paper lists and 1.35 / 0.87 / 2.94 / 12.6 at the deepest fitted pairmax points of Section 6 (layout order as in the table).

bins are close in distance everywhere and keep a large density cross. Large magnification over small density is the DES far-pair regime (Section 4.3), reached here by widening the bins rather than separating them.

The scale cuts filter part of it. Most of the pairmax signal on $\Delta z = 0.2$ falls outside the ultraviolet-fraction scale cuts (at $k_{\rm max} = 0.2$ its pairmax total drops from 63.1 to 12.6). That is not an accident of the physics but of the cut's construction: the cut was built on the magnification-free model, and a pair whose density cross spectrum is tiny keeps few bands, however much magnification signal those bands would carry. A cut rebuilt with magnification on would keep different bands; that rebuild is an open issue (Appendix A).

The record of the external estimate. Before the code extension, an external Limber evaluation with top-hat bins of the $\Delta z = 0.02$ layout estimated $\Delta\chi^2 \sim 0.01$–0.05 on that layout's paper list and ~ 0.3–1 on its pairmax list. The pipeline values are about ten times larger on the thin layouts (0.48 and 1.44), and the estimate did not cover $\Delta z = 0.05$ and $\Delta z = 0.2$ at all. The estimate's *relative* anatomy – adjacent pairs weakest, added pairs stronger, far separations strongest – survives; its absolute normalization did not, which is why every conclusion in this note rests on the pipeline numbers.

Pending – The data figures `mag_kernels` (one thin bin's density kernel against its magnification kernel) and `mag_pair_terms` (the band-level terms of one adjacent and one $g = 2$ pair against the band errors), from the extended code.

5 What cosmolike computes

This section records the state of the code on branch `lowrank_fftlog` as audited on 2026-11-13, so the note's statements are checkable line by line.

5.1 The kernel and the coefficient

Cosmolike implements the DES Y3 model of Section 3.6. The kernel (`radial_weights.c:97`) is $W_{\rm mag}(a, f_K, n_z) = 1.5\,\Omega_m(f_K/a)\,g_{\rm lens}(a, n_z)$ in units where $H_0/c = 1$, with $g_{\rm lens}$ (`redshift_spline.c:2578`) the bin-averaged efficiency of Eq. (10) built from the lens $n(z)$ – in this project the Fingers-of-God-convolved $n(z)$, the same table the density kernel reads, which is consistent (the blur approximates the true radial positions of the bin's galaxies) and changes the broad kernel by only $\mathcal{O}(10^{-3})$ of itself. The coefficient (`bias.c:166`, `gbmag`) is one constant per lens bin, applied by the callers. For 55–60 Mpc bins the within-bin evolution of the count slope is negligible, so *the per-bin constant is an exact sampling of $s(z)$ at up to ninety points*: the approximation that costs DES-like wide bins up to $\sim 1\sigma$ at Stage IV [13] costs a thin-bin survey nothing. This project's likelihood currently passes zero for every bin (`likelihood/desi_cl.py`, `set_lens_related`).

5.2 The $\ell(\ell+1)/(\ell+1/2)^2$ prefactor, derived

The convergence is not an independent field: it is the transverse Laplacian of the lensing potential ψ (the projected gravitational potential), $\kappa = \frac{1}{2}\nabla^2_\theta\psi$. On the sphere, the Laplacian of a multipole ℓ multiplies it by $-\ell(\ell+1)$, so $\kappa_{\ell m} = -\frac{1}{2}\,\ell(\ell+1)\,\psi_{\ell m}$: the *exact* ℓ dependence (the sign squares away in the power spectrum). The Limber evaluation, meanwhile, converts the potential to the density through Poisson's equation with the *transverse* wavenumber $k = (\ell+1/2)/\chi$, which brings a factor $1/k^2\chi^2 = 1/(\ell+1/2)^2$. The product is the prefactor cosmolike multiplies into every magnification term (`cosmo2D.c:4089, 5616`):

$$\frac{\ell(\ell+1)}{(\ell+1/2)^2} \;=\; 1 - \frac{1}{4(\ell+1/2)^2} \;\to\; 1 \quad \text{at large } \ell. \tag{12}$$

It is the exact spherical correction, not an approximation; at $\ell = 15$, the survey's first band, it is 0.9990.

5.3 Where the terms live: Limber, FKEM, low-rank

In the Limber C_ℓ^{gg} the integrand squares the summed kernel $(W_{\rm gal}b_1 + W_{\rm mag}\,\epsilon_\ell\,C_i + W_{\rm RSD})$ (`cosmo2D.c:6029`, ϵ_ℓ the prefactor above), so all three products are kept: density against magnification, the magnification auto term, and magnification against RSD. In the non-Limber paths, the FKEM method – the generalized-FFTLog evaluation of the exact spherical-Bessel projection [6] – carries a magnification row exactly at linear level (growth factorized) and hands its nonlinear correction to Limber; the low-rank engine replaces only the density×density block and leaves the magnification row on the FKEM path (`cosmo2D.c:68--78`). The magnification leg multiplies the nonlinear *matter* spectrum with the linear galaxy bias ($b_1 P_{\rm mm}$): the one-loop galaxy terms stay on the density×density block, the DES Y3 "tree-level lensing leg" convention [4] – crossing loop bias with the lensing leg would be formally two-loop, and here it would correct a term that is itself $\lesssim 10^{-3}$ of the signal.

5.4 What is omitted

The complete linear theory of observed number counts [1, 2, 3] contains, beyond density, RSD and magnification: *Doppler terms* (the galaxies' peculiar velocities enter the observed

redshift and flux beyond the Kaiser compression already modelled) and *potential terms* (Sachs–Wolfe-like contributions of the gravitational potentials). Cosmolike carries none of them. The potential terms are orders of magnitude below everything at these scales. The Doppler terms scale as v/c times factors like $(2-5s)\,c/(\chi\,aH)$, which grow toward low redshift as χ shrinks: the place to look is the lowest-z BGS adjacent pairs, where $c/(\chi aH) \approx 7$ at $z = 0.15$, and the largely antisymmetric (in bin exchange) shape is partially filtered by the symmetric band powers.

Pending – T6: the one-paragraph Doppler estimate – the evaluated size of the leading Doppler term on the lowest-z BGS adjacent pairs, against the band errors, from the linear-counts expressions of arXiv:1105.5292.

5.5 The covariance

The covariance keeps $C_i = 0$ (decided 2026-11-13): with coefficients of order 3 the covariance's own magnification terms would change the cross-spectrum variances at the 10^{-3} level, far below the covariance's other stated approximations, and DES Y3 made the same choice for its fixed-coefficient analysis [4].

5.6 The cross-pair extension: what was done and validated

The historical restriction – the gg cross-pair Limber batch was fatal when either bin of a pair carried a nonzero coefficient (`cosmo2D.c:13115`), because the pair quadrature covered only the two bins' supports, not the foreground where the magnification kernel lives – has been lifted on branch `lowrank_fftlog`. What the pair batch does now:

- Each bin's kernel gains the magnification row $W_{\rm mag}\,\epsilon_\ell\,C_b$, with $\epsilon_\ell = \ell(\ell+1)/(\ell+1/2)^2$ – the same row the auto Limber work squares. The product of the two bins' kernels then carries every term: density against magnification in both orderings, the magnification auto term, and magnification against RSD.
- The magnification rows ride the nonlinear matter power, and the one-loop terms stay on the density×density part: the auto work's tree-level convention, unchanged. The α_2 counterterm keeps the kernel without magnification, as the auto and FFTLog counterterms do.
- The pair quadrature keeps its panels at the two bins' padded $n(z)$ edges, so the density terms' nodes never depend on the coefficients; when both bins carry a coefficient, one additional panel covers the foreground – from the nearer support edge to the source-sample edge – for the magnification auto term. The FKEM and low-rank pair cores already had their magnification slots, so the non-Limber linear part is unchanged.

The likelihood gains one method, `set_magnification_coefficients` in `likelihood/desi_cl.py`: one $C = 5s - 2$ per bias group, expanded to a per-bin `b_mag`, default zero. It is deliberately not a configuration-file option: a new option would change the resolved input whose checksum every archived derivative cache records, and every cache would stop matching its model. The δd caches and the shift fits are written by `scripts/compute_magnification_vectors.py` and `scripts/compute_magnification_fits.py`.

Validation (all passed). (i) $C = 0$ is bitwise: the fiducial vector equals the derivative cache's fiducial exactly, on all eight layout–list combinations, and again after a coefficient set-and-reset. (ii) Polynomial response: the vector is linear in C plus the quadratic magnification auto term; the four-point identity $d(2C) - 3d(C) - d(-C) + 3d(0) = 0$ holds to about 10^{-5} of δd on the cross pairs and to $\leq 5 \times 10^{-3}$ of δd on the autos (the autos' δd is itself $\leq 1\%$ of the auto spectrum; the residual is a numerical effect of the auto non-Limber rows, not yet traced – an open issue). (iii) Cross equals auto: giving bin j of a pair an exact copy of bin i's $n(z)$, biases and coefficient makes the cross spectrum C_ℓ^{ij} equal the auto spectrum C_ℓ^{ii} to 3.8×10^{-5} with magnification on – the same level as with it off – so the pair code's magnification terms agree with the autos' to about 3×10^{-4} of their size. (iv) The paper-list point fitted inside the pairmax system reproduces the notebook's paper-list fit to every printed digit, and the pairmax FoM gains reproduce the pair-axes figure of `notes/cross_correlations` exactly. *Not yet done:* the brute-force spherical-Bessel check of the magnification rows (the "Limber-safe" nonlinear hand-off stays an assumption) and the Doppler estimate (Section 5.4); both are open issues (Appendix A).

6 The shift test

The what-if framing. A Fisher forecast's synthetic data are the fiducial model's own prediction, so the forecast itself has no shift: model and data agree at the fiducial by construction. The shift test asks a what-if question: *if nature carries magnification at the measured coefficients and the model leaves it out, where does the fit land, in units of its own error ellipse?* The answer is a safety estimate for the future MCMC analyses, not a property of the published forecast.

The linear-response method. Let $m(\theta)$ be the model data vector, D its derivative matrix at the fiducial θ_0 (the derivative bases the project already caches), C the data covariance, and $F = D^T\mathsf{C}^{-1}D$ (plus priors) the Fisher matrix. Give the data the magnification part, $d = m(\theta_0)+\delta d$ with δd the difference between the data vector with and without the coefficients of Table 1. A Gaussian fit of the magnification-free model to those data lands, to first order, at

$$\delta\theta \;=\; F^{-1}D^T\mathsf{C}^{-1}\,\delta d\,: \tag{13}$$

the best-fit displacement is the excess signal projected onto the model's parameter directions. The derivation, its assumptions and its validation against nonlinear refits are in `notes/fisher_machinery`; this note reuses the machinery unchanged, with δd the only new ingredient. The same linear-response anatomy, applied to neglected magnification in a 3×2pt forecast, is worked through in [12].

What is reported. The shifts of w_0, w_a, Ω_m and $10^9 A_s$ in units of each parameter's marginalized error, and the (w_0, w_a) shift in units of its own error ellipse, $r = [\delta\theta_2^T\,\mathsf{C}_2^{-1}\,\delta\theta_2]^{1/2}$ with C_2 the pair's 2×2 marginal covariance – as MCMC-safety estimates, with no pass/fail threshold. "Free" means every nuisance parameter varies; "frozen" holds the nuisances at the fiducial (the cosmology-only block), each case in its own errors. The frozen companion separates what the nuisance freedom – above all the per-group linear-bias level and slope – absorbs from what lands on cosmology. The δd comes from the extended code of Section 5.6, so it

layout, list	free: $r(w_0, w_a)$ / Ω_m / A_s $C_{\rm ELG} = 0$	$+2$	-2	frozen r at 0
62 Mpc paper	0.05/0.03/0.05	0.06/0.03/0.07	0.21/0.09/0.15	0.13
62 Mpc pairmax ($g \leq 3$)	0.07/0.03/0.06	0.36/0.14/0.27	0.15/0.07/0.11	0.12
$\Delta z = 0.02$ paper	0.05/0.03/0.05	0.01/0.02/0.02	0.24/0.11/0.19	0.11
$\Delta z = 0.02$ pairmax ($g \leq 2$)	0.06/0.03/0.06	0.30/0.12/0.23	0.09/0.07/0.10	0.11
$\Delta z = 0.05$ paper	0.05/0.03/0.04	0.05/0.06/0.12	0.19/0.14/0.19	0.17
$\Delta z = 0.05$ pairmax ($g \leq 3$)	0.05/0.06/0.05	0.46/0.30/0.39	0.27/0.23/0.23	0.15
$\Delta z = 0.2$ paper	0.19/0.16/0.10	0.29/0.21/1.00	0.31/0.30/0.27	0.30
$\Delta z = 0.2$ pairmax ($g \leq 2$)	0.19/0.16/0.10	0.38/0.28/1.06	0.30/0.27/0.27	0.30

Table 4: The shift test's results: the largest shift over the cuts $k_{\rm max} \leq 0.2\,h/{\rm Mpc}$ (at $z = 0$), per layout and pair list. "Free" columns: the (w_0, w_a) ellipse shift r and the absolute shifts of Ω_m and $10^9 A_s$ in units of their marginalized errors, with every nuisance free, at the fiducial coefficients ($C_{\rm ELG} = 0$) and the two ELG bracket corners. The last column is the frozen (cosmology-only) r at the fiducial coefficients. "62 Mpc" abbreviates $\Delta d_{\rm com} = 62$ Mpc. All values are what-if MCMC-safety estimates; no threshold is attached.

carries the blurred $n(z)$, the real bands, masks and scale cuts, and the same kernels as the derivative bases; the shifts were computed at the final derivative tier, per cut $k_{\rm max} \leq 0.2\,h/{\rm Mpc}$ (at $z = 0$), on every layout and both pair lists, at the fiducial coefficients and the ELG bracket corners. One bookkeeping detail: the fitted points walk the gap axis, so on $\Delta z = 0.02$ the deepest point keeps the pairmax list's $g \leq 2$ pairs (363 of the 370); the seven σ_v-union pairs of that list enter the $\Delta\chi^2$ totals of Table 3 but no fitted point. The signed per-cut values live in the run records (`chains/magnification_shifts_*.json`); the model is the fiducial model (one-loop, $\alpha_2 = 0$, σ_v prior widths $\times 6$, the pessimistic linear-bias prior).

Results. Table 4 gives the largest value over the cuts; Figure 3 shows the per-cut values with every nuisance free.

What the numbers say. *At the measured coefficients* ($C_{\rm ELG} = 0$), leaving magnification out moves the thin-layout fits by at most about 0.07 of the error, on both pair lists; adding the pairmax pairs barely changes the shifts, because magnification across two to four thin-bin widths is still small (Section 4.2). *The unmeasured ELG count slope dominates.* At $C_{\rm ELG} = -2$ the paper lists reach 0.2–0.3; at $C_{\rm ELG} = +2$ the pairmax lists reach 0.30–0.46 against at most 0.06 on the thin paper lists. In that corner the free nuisances make the shift *larger*, not smaller (on $\Delta z = 0.05$ pairmax: free 0.46, frozen 0.20): the added pairs' magnification reads as a change of the per-group linear-bias level and slope, exactly the freedom whose extra constraint produces the pairmax FoM gain (`notes/cross_correlations`). The gain and the magnification sensitivity are two faces of the same channel. *At the fiducial coefficients the nuisance freedom absorbs most of the signal*: holding the nuisances fixed raises the (w_0, w_a) shift by a factor 1.5 to 13 cut by cut (more where the free shift nearly vanishes); at the maxima over cuts the factor is 1.7 to 3.5 on the thin layouts and 1.6 on $\Delta z = 0.2$. $\Delta z = 0.2$ *is different in kind*: its cross spectra are magnification dominated (Section 4.5), it moves 0.19 even at the fiducial coefficients, and its A_s shifts by about 1σ at $C_{\rm ELG} = +2$. Its FoM is the smallest of the four layouts, so this matters for the wide-bin comparison rather than for the headline

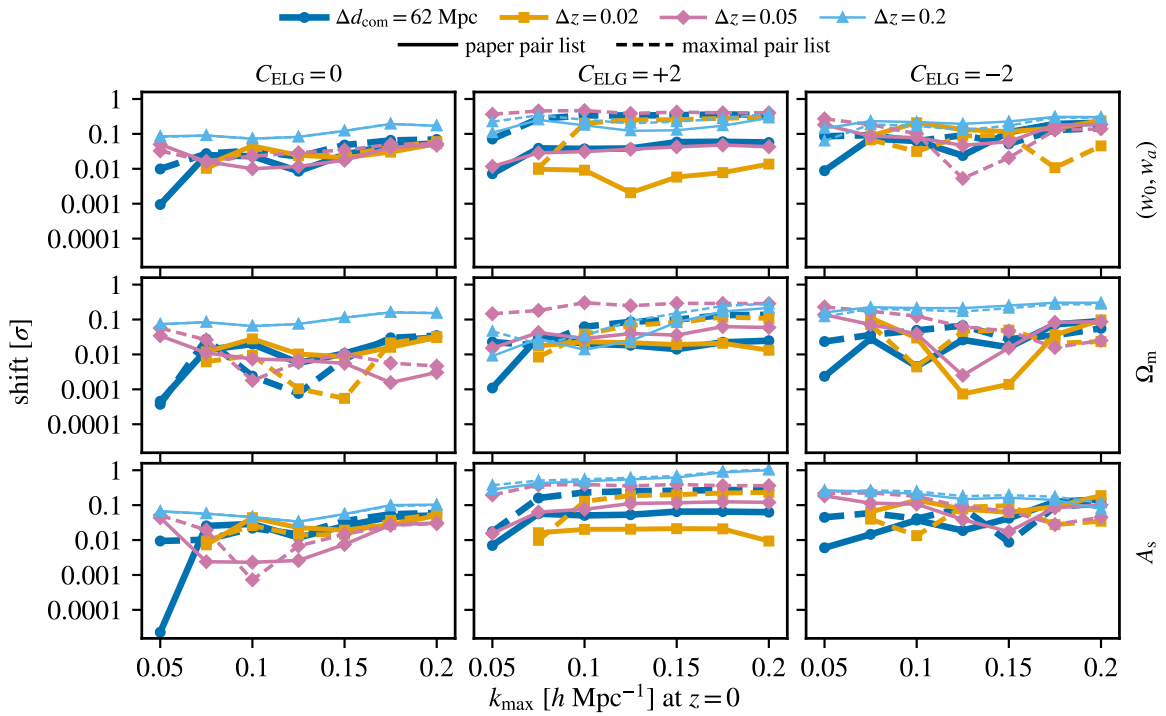


Figure 3: The shift test against the scale cut: what-if MCMC-safety estimates of where the magnification-free fit lands if nature carries magnification at the stated coefficients. Rows: the (w_0, w_a) shift in units of its error ellipse, and the absolute shifts of Ω_m and A_s in units of their marginalized errors. Columns: $C_{\rm ELG} = 0$, $+2$ and -2, with BGS $C = 2$ and LRG $C = 3$ throughout. The four layouts are distinguished by colour, line width and marker; solid lines are the paper pair lists, dashed lines the maximal (pairmax) lists. Every nuisance is free; the frozen (cosmology-only) values are in the text and the run records. The shared vertical axis is logarithmic, the quoted $k_{\rm max}$ is the $z = 0$ value of the per-bin cut at $k_{\rm max}/D(\bar{z}_i)$, and no threshold line is drawn.

forecasts.

7 Magnification in the model

If the shift-test numbers are judged too large to leave the term out – or if the pair list is ever extended toward genuinely far pairs – the coefficients move into the model. The method is fixed by the structure of the signal; only the results wait.

Exact template columns. The data vector is linear in each C_i through the terms that cross density with magnification, and quadratic through the magnification auto terms; nothing else depends on the coefficients. Each coefficient's derivative column is therefore an exact template – a handful of evaluations per group isolates the linear and quadratic pieces algebraically, with no stencil noise – exactly as the project already treats the stochastic amplitudes. With one coefficient per sample group there are six new columns.

The cosmology columns change. The magnification terms carry Ω_m, the growth factor and the distances, so every cosmology derivative must be recomputed with the coefficients on: the cached magnification-free derivative bases serve the shift test, not the extended forecast. The runtime cost (the magnification kernels widen every lens bin's quadrature support to the full foreground) is measured before this step is scheduled.

Coefficient treatments. Three treatments, in increasing order of caution: fixed at the measured values (DES Y3's baseline, justified there by the fixed-versus-Gaussian difference being 0.06–0.19σ in S_8 [5]); Gaussian priors at the measurement widths; free. With far pairs in the data vector the free treatment becomes self-calibrating – the cross spectra measure the coefficients [5, 17, 10] – and magnification turns from a nuisance into a signal.

Pending – T3 results: FoM(w_0, w_a) and FoM($\Omega_m, 10^9 A_s$) against $k_{\max}$ with the coefficients off / fixed / Gaussian / free, paper and pairmax lists (figure `mag_fom_cost`); and, if the far-pair extension T4 runs, $\sigma(C_g)$ against the pair reach (figure `mag_selfcal`). Before the exact-template derivative columns are built, the autos' 5×10^{-3} polynomial-response residual (Section 5.6) must be traced.

A Open issues

- **The ELG count slope is unmeasured.** Table 1 brackets $C_{\rm ELG}$ in $[-2, +2]$; the finite-difference measurement on the DESI ELG target selection (with the fibre-magnitude response, Section 3.7) replaces the bracket when it runs.
- **The "Limber-safe" hand-off is untested for thin pairs.** The FKEM magnification row is exact at linear level; its nonlinear correction rides Limber on the assertion that the broad kernel is safe (`cosmo2D.c:76--78`). One brute-force spherical-Bessel check on a thin pair settles it (Section 5.6, the one validation not yet run).
- **The autos' polynomial-response residual.** The four-point linearity identity closes to 10^{-5} of δd on the cross pairs but only to 5×10^{-3} on the autos (Section 5.6); the residual sits in the auto non-Limber rows and must be traced before the exact-template coefficient derivatives of Section 7 are built.
- **The scale cuts were built without magnification.** On $\Delta z = 0.2$ most of the pairmax magnification signal falls outside the ultraviolet-fraction cuts (Section 4.5); a cut rebuilt on the model with magnification would keep different bands.
- **Doppler terms at low redshift.** The leading omitted count term; the estimate for the lowest-z BGS adjacent pairs is pending (Section 5.4).
- **Far pairs.** Beyond the pairmax lists magnification is both mandatory in the model and a measurable signal; whether the data vector is ever extended that far is a separate decision, with the self-calibration forecast (T4) as its input.

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